

CYBERSECURITY & ETHICAL HACKING

WEEK 3 — PROJECT MODULE 1

PASSWORD CRACKING WITH JOHN THE RIPPER (JTR)

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Program / Batch	B083F
Date	25/09/2026
Project	W3-PM1 — Password Cracking with JTR
Platform	Windows PC
Tools	John the Ripper (John) and Johny GUI

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1. Executive Summary

This report documents the practical completion of Week 3 Project Module 1: Password Cracking with John the Ripper (JTR). The supplied task requires the learner to use JTR John and the JTR Johny graphical interface on a Windows PC to recover the password of an encrypted PDF in an authorized training environment. The exercise demonstrates the workflow of extracting a PDF password hash, saving the hash in a text file, loading it into Johny, starting an attack, and using the recovered password to open the protected PDF.

2. Task Overview

The source task identifies John the Ripper as a password-cracking tool used by security professionals to test password strength. It also identifies Johny as the graphical version of John the Ripper. The practical objective is to crack the password of the attached protected PDF using JTR John and JTR Johny on a Windows PC.

3. Objectives

- Install or prepare John the Ripper and the Johny GUI.
- Configure Johny to use the John.exe executable.
- Obtain the hash representation of the encrypted training PDF.
- Save the extracted hash in a text file in the required format.
- Load the hash file into Johny.
- Start a password-cracking attack and document the result.
- Use the recovered password to open the protected PDF.
- Provide screenshot evidence for each major stage.

4. Tools and Resources

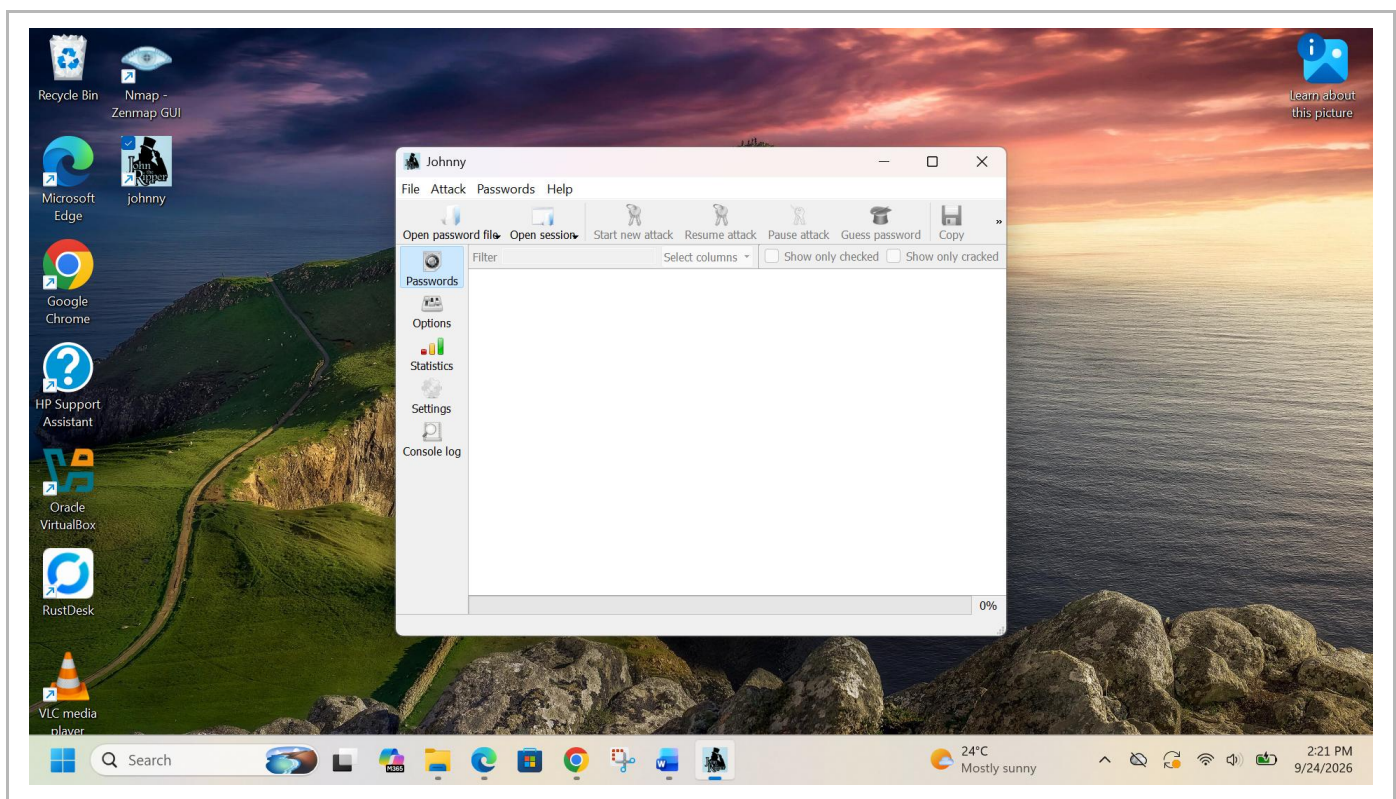
The source material specifies John the Ripper, Johnny GUI, an encrypted PDF file, a text editor such as Notepad, and an online PDF hash extraction tool. The supplied instructions provide Openwall download locations for John the Ripper(<https://www.openwall.com/john/>) and Johnny(<https://openwall.info/wiki/john/johnny>), and provide the PDF hash extractor URL (<https://www.onlinehashcrack.com/tools-pdf-hash-extractor.php>).

Tool / Resource	Purpose	Evidence
John the Ripper (JTR)	Password/hash cracking engine	[Screenshot]
Johnny GUI	Graphical interface for JTR	[Screenshot]
Encrypted training PDF	Target file supplied for the lab	[Screenshot]
Notepad / text editor	Store extracted hash in hash1.txt	[Screenshot]
PDF hash extractor	Extract PDF hash for use with JTR	[Screenshot]

5. Methodology and Procedure

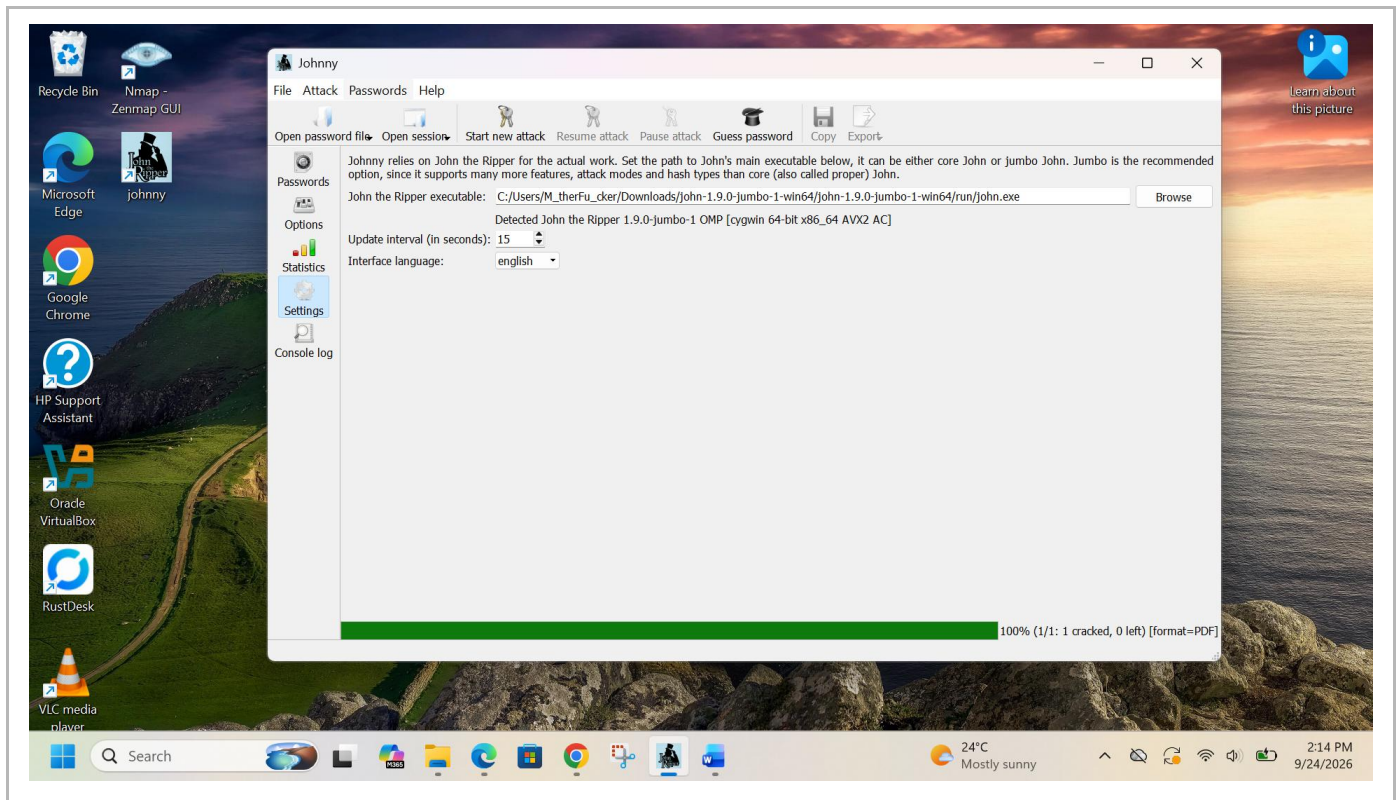
5.1 Install / Prepare John the Ripper and Johnny

The source instructions direct the learner to download John the Ripper from an Openwall source and to download Johnny from its official Openwall information/download location. After installation, Johnny is opened and configured through Settings by browsing to the John.exe executable located in the JTR run folder.



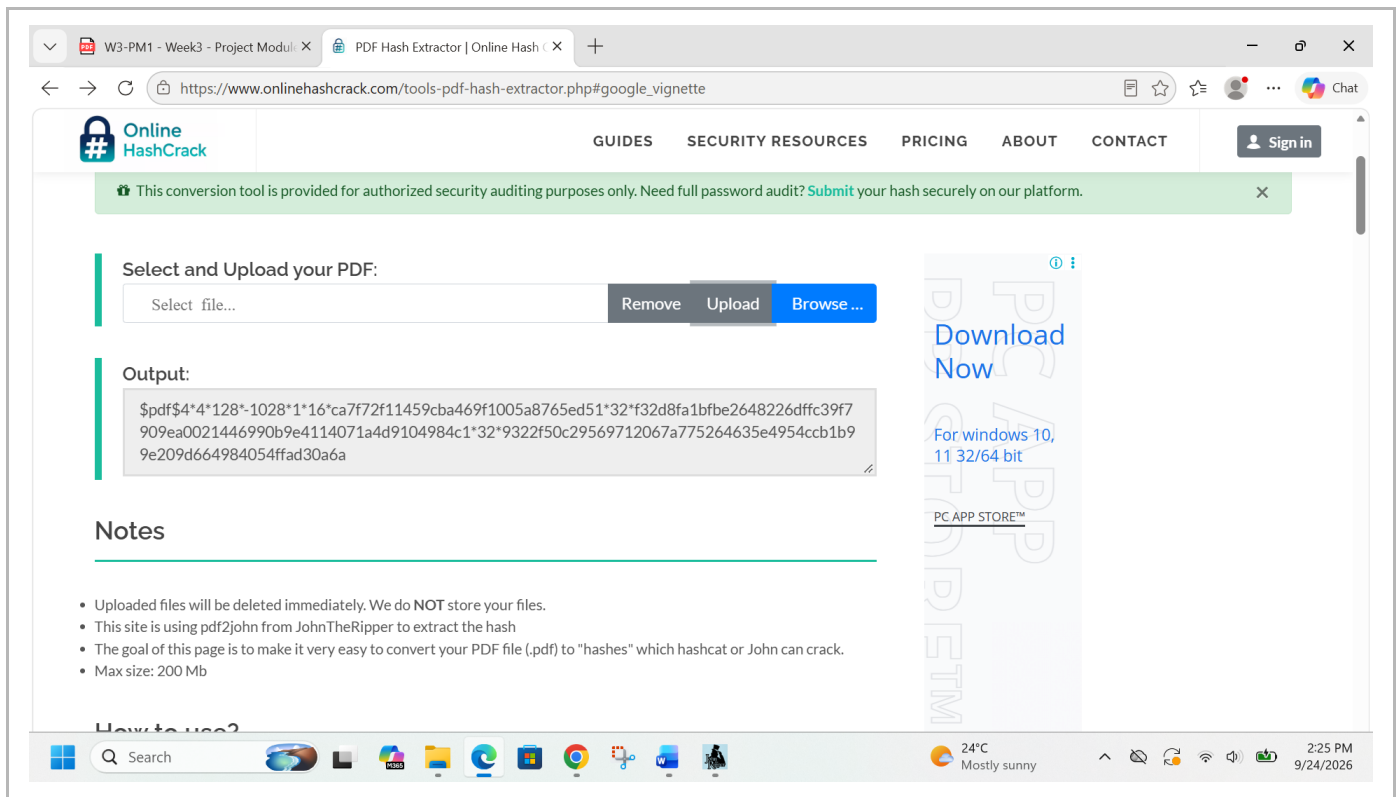
5.2 Configure Johnny to Use John.exe

Open Johnny, select Settings, and browse to the John.exe file in the JTR run folder. Confirm that the correct executable is selected before proceeding.



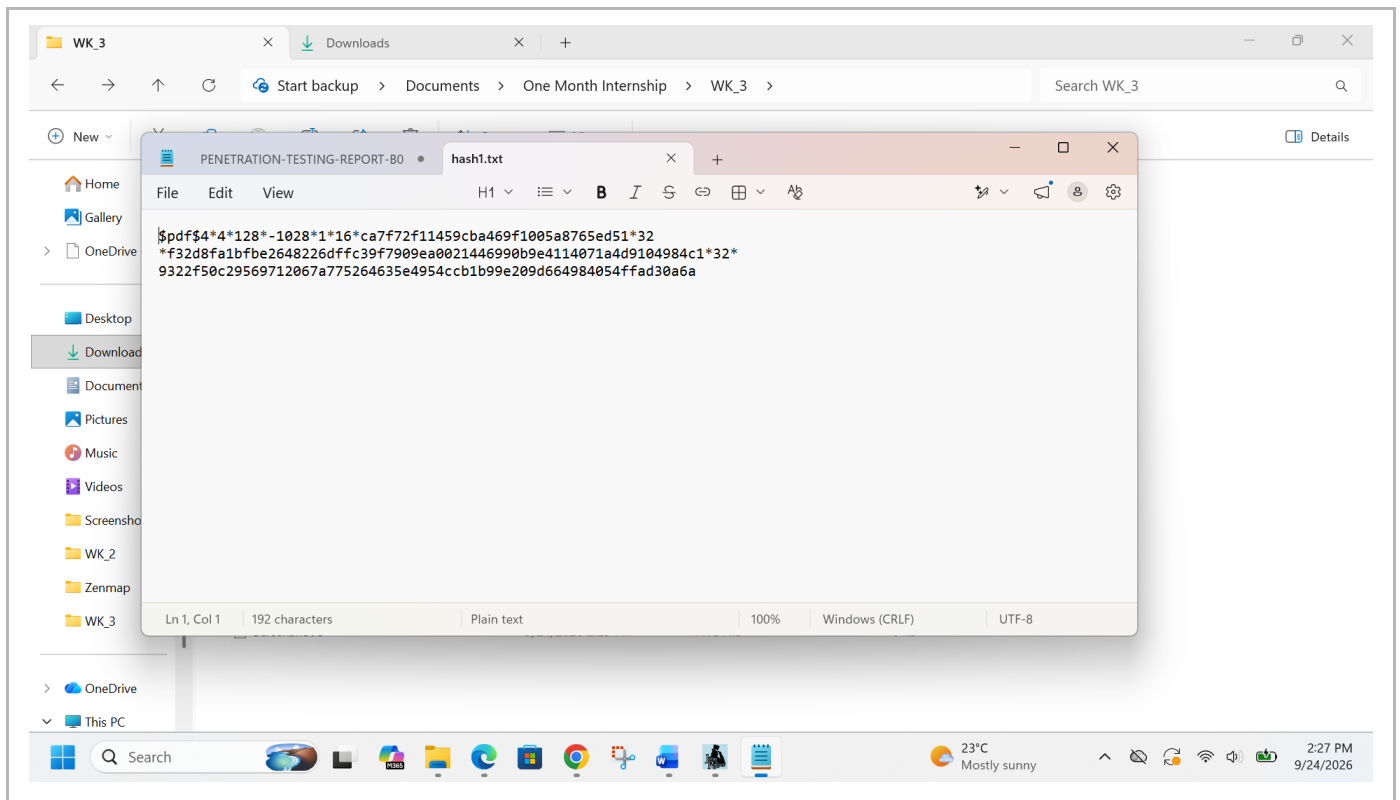
5.3 Obtain the PDF Hash

Download the encrypted training PDF and use the specified PDF hash extraction process. The source instructs the learner to upload the PDF to the listed PDF hash extractor, then copy the resulting hash value.



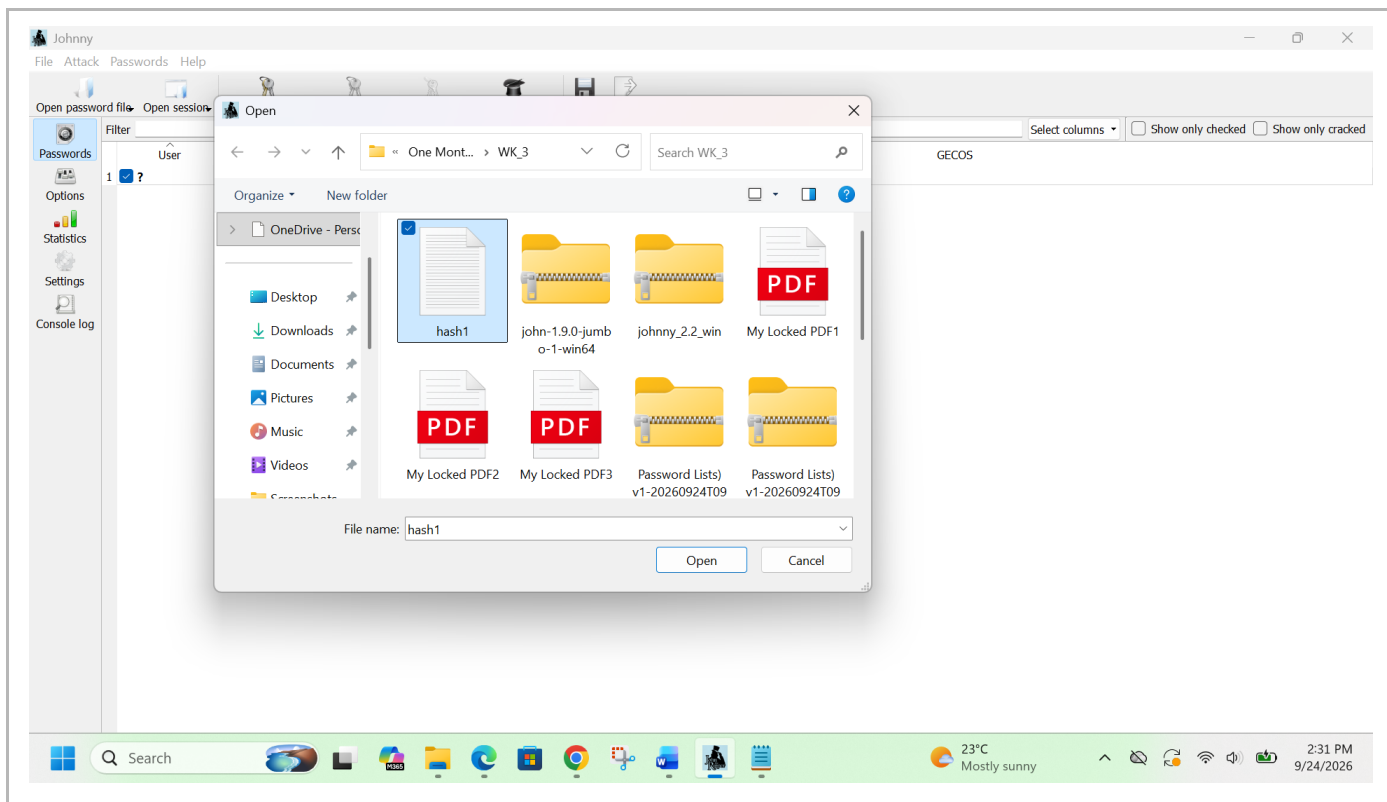
5.4 Create hash1.txt

Open Notepad, paste the extracted hash, and save it as hash1.txt. The source specifically notes that extra characters such as b' at the beginning should be removed and that the hash should be saved in the expected \$pdf\$... format.



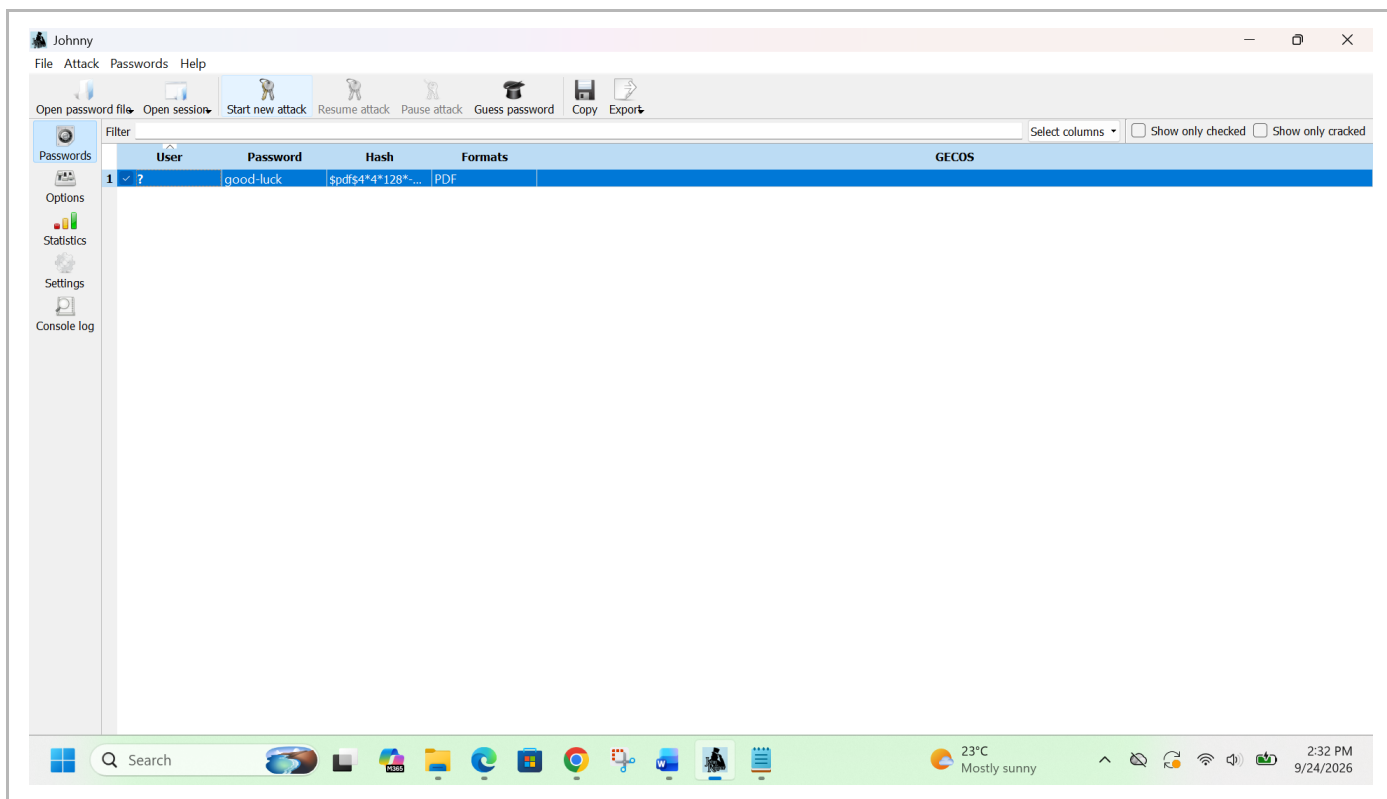
5.5 Load the Hash File into Johnny

Open Johnny and select 'Open password file'. Browse to the hash1.txt file created in the previous step and open it.



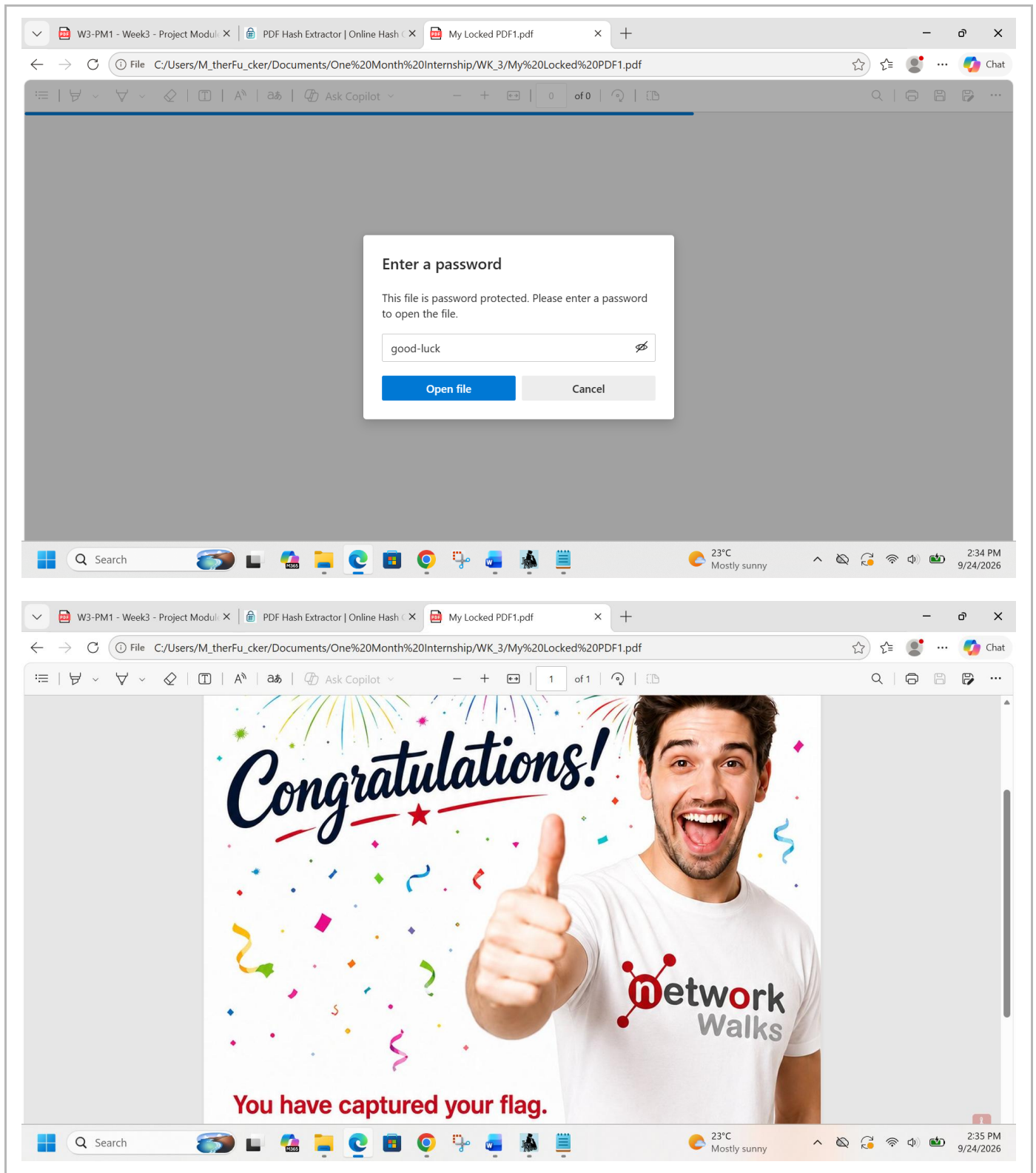
5.6 Start the Password-Cracking Attack

Select 'Start new attack'. The source notes that the cracking process can take time depending on computer performance and password complexity.



5.7 Verify the Recovered Password

After the password is recovered, use it to open the encrypted PDF. The source demonstrates entering the cracked password and confirming that the PDF opens successfully.



6. Results

Item	Learner Result
JTR installed / available	Completed
Johnny configured with John.exe	Completed
PDF hash extracted	Completed
hash1.txt created	Completed
Attack completed	Completed
PDF opened successfully	Completed

7. Evidence Log

Evidence #	Activity	Screenshot Added?
1	JTR / Johnny installation	Yes
2	John.exe configuration	Yes
3	PDF hash extraction	Yes
4	hash1.txt creation	Yes
5	Hash loaded into Johnny	Yes
6	Attack started / completed	Yes
7	PDF opened successfully	Yes

8. Security and Ethical Considerations

Password-cracking tools are dual-use security tools. This practical exercise should only be performed against files, systems, or accounts for which the learner has explicit authorization. In this project, the target is the supplied training PDF. The objective is to understand password security and the mechanics of a controlled password-recovery exercise, not to gain unauthorized access to third-party data.

9. Challenges / Observations

Use this section to record what happened during your own practical session.

Installation / configuration issue: **None**

Hash extraction issue: **None**

Cracking time observed: **No but was not more than 5 minutes after everything was set.**

Password complexity observation: **Moderate complexity.**

10. Conclusion

The practical exercise demonstrates a complete password-recovery workflow using John the Ripper and the Johnny GUI: preparing the tools, extracting a PDF hash, saving the hash to a text file, loading the hash into Johnny, initiating an attack, and verifying the recovered password by opening the protected PDF. The

supplied task states that the exercise is intended to help learners understand how password cracking works and why strong passwords are important for protection.

11. Reference

Networkwalks, 'Week 3 | Project Module 1 — Password Cracking with JTR', supplied project task PDF.

PASSWORD CRACKING WITH NETWORKWALKS TOOLS

Week 3 | Project Module 2

1. Executive Summary

This report documents the completion of the Week 3 Project Module 2 practical task, **Password Cracking with Networkwalks Tools**. The task requires a locked PDF to be processed using the Networkwalks Hash Calculator to extract a PDF password hash, followed by the Networkwalks Password Cracker to identify the corresponding password. The final stage is to use the recovered password to open the protected PDF.

The practical is presented as a cybersecurity and ethical-hacking exercise designed to demonstrate how weak passwords can be recovered and why strong password practices are important. The source task specifies that both tools operate through a web browser and can therefore be used on a Windows laptop or Kali Linux environment.

2. Task Overview

Task	Crack the password of the supplied locked PDF using the Networkwalks Hash Calculator and Password Cracker.
Operating Environment	Windows laptop (the task also notes that Kali Linux may be used).
Primary Tools	Networkwalks Hash Calculator and Networkwalks Password Cracker.
Expected Outcome	Recover the PDF password and successfully open the protected PDF.

3. Background and Technical Context

According to the supplied lab instructions, password cracking is the process of recovering a password from stored data or a protected file. The lab explains that a protected file can contain password-related information in hash form. The workflow therefore begins by extracting the relevant hash from the PDF and then testing candidate passwords against that hash until a match is found.

The exercise specifically uses two browser-based Networkwalks tools. The Hash Calculator extracts a PDF hash beginning with “\$pdf\$...”, while the Password Cracker accepts the extracted hash and attempts passwords until a matching value is identified.

4. Objectives

- Obtain the encrypted PDF specified by the practical task.
- Use the Networkwalks Hash Calculator to extract the PDF hash.
- Copy the complete hash value beginning with \$pdf\$.
- Submit the hash to the Networkwalks Password Cracker and start the cracking process.
- Record the recovered password displayed by the tool.
- Use the recovered password to open the locked PDF and confirm successful completion.

5. Tools and Resources

Resource	Purpose	URL / Location
Networkwalks project task page	Source of the practical task and encrypted PDF.	networkwalks.com/project-task-lab-password-cracking-with-networkwalks-tools/
Networkwalks Hash Calculator	Extracts the password-protected PDF hash.	networkwalks.com/hash-calculator/
Networkwalks Password Cracker	Tests candidate passwords against the supplied hash.	networkwalks.com/password-cracker/
Adobe Acrobat Reader / PDF reader	Used to open the protected PDF after password recovery.	Installed PDF reader on Windows

6. Methodology and Procedure

The following procedure follows the sequence provided in the Networkwalks practical document. The screenshot spaces are intentionally left editable so that personal evidence of completion can be inserted into this report.

STEP 1 | Obtain the encrypted PDF

Download the encrypted PDF file identified in the task instructions as “My Locked PDF1.pdf” from the Networkwalks project task page.

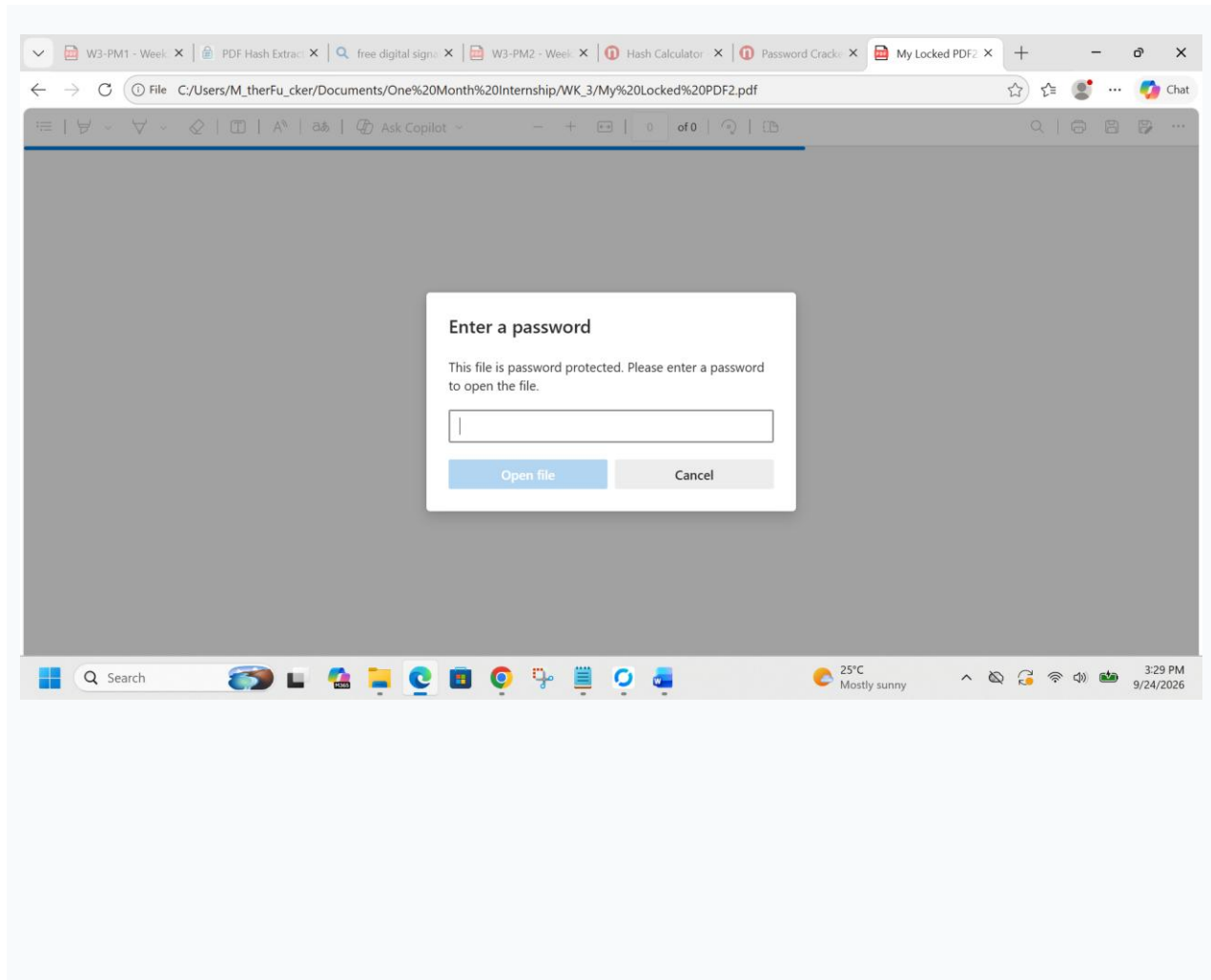


Figure 1. Encrypted PDF obtained.

STEP 2 | Open the Hash Calculator

Open the Networkwalks Hash Calculator in a web browser. The task documentation provides the Hash Calculator URL(<https://networkwalks.com/hash-calculator/>) and shows the browser-based interface used for the exercise.

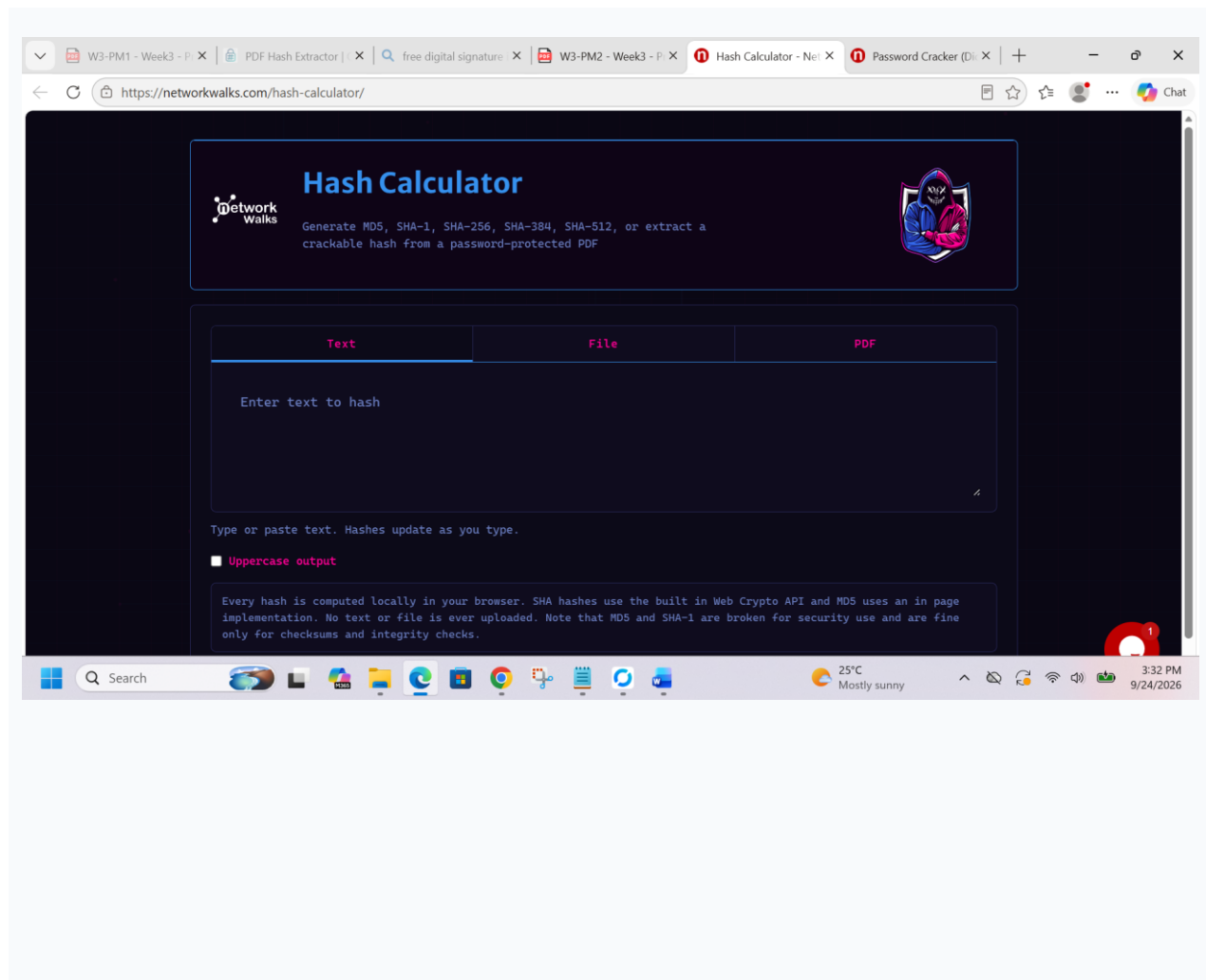


Figure 2. Networkwalks Hash Calculator.

STEP 3 | Upload the locked PDF and extract the hash

Upload the locked PDF to the Hash Calculator. According to the task instructions, the tool reads the file and produces a hash value beginning with "\$pdf\$...".

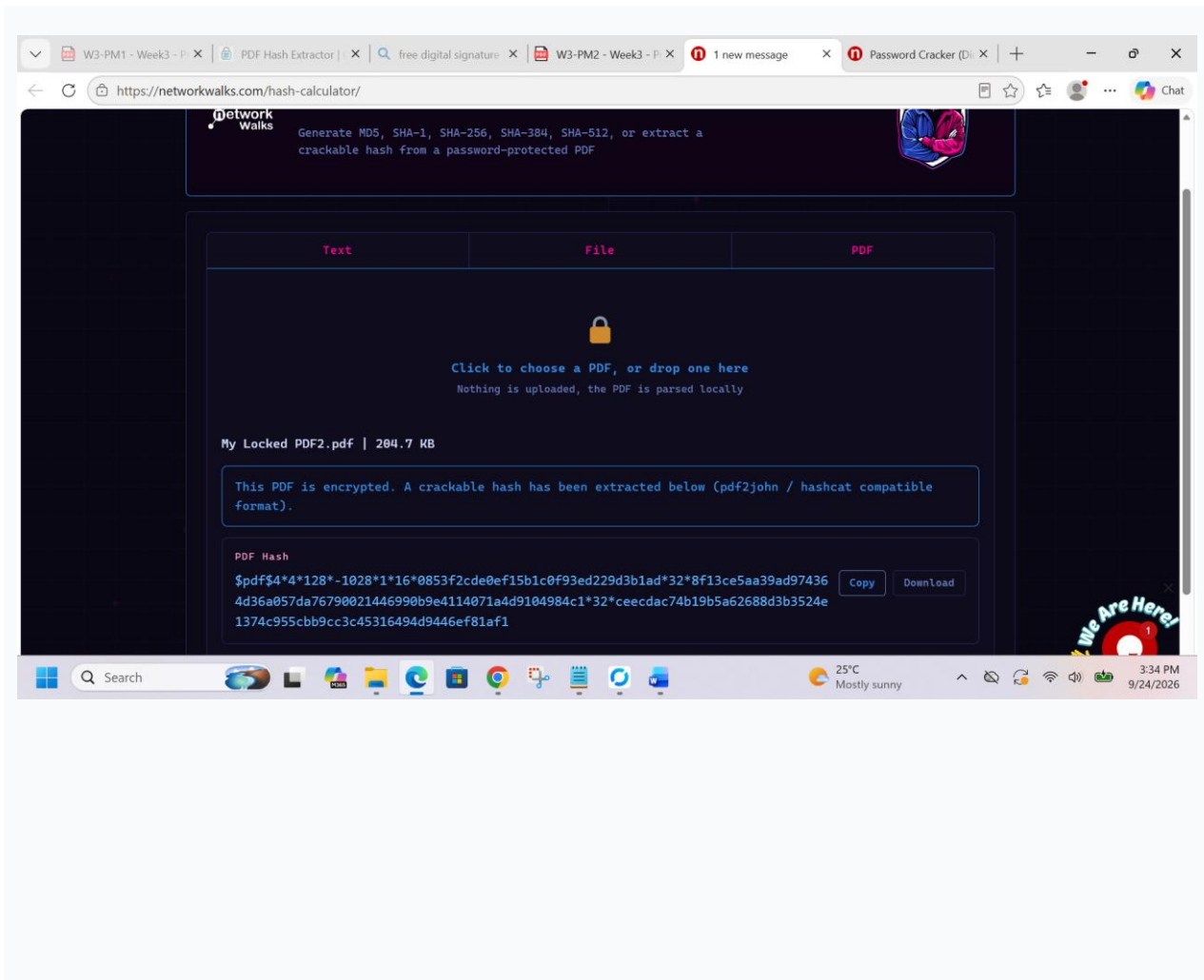


Figure 3. PDF hash extracted.

STEP 4 | Copy the complete hash

Copy the full hash value generated by the Hash Calculator. The practical specifically warns that the complete value must be copied, including the initial “\$pdf\$” portion, without omitting any characters.



Figure 4. Complete hash copied.

STEP 5 | Open the Password Cracker

Open the Networkwalks Password Cracker in a web browser (<https://networkwalks.com/password-cracker/>). The supplied task document identifies this as the second Networkwalks tool required for the practical.

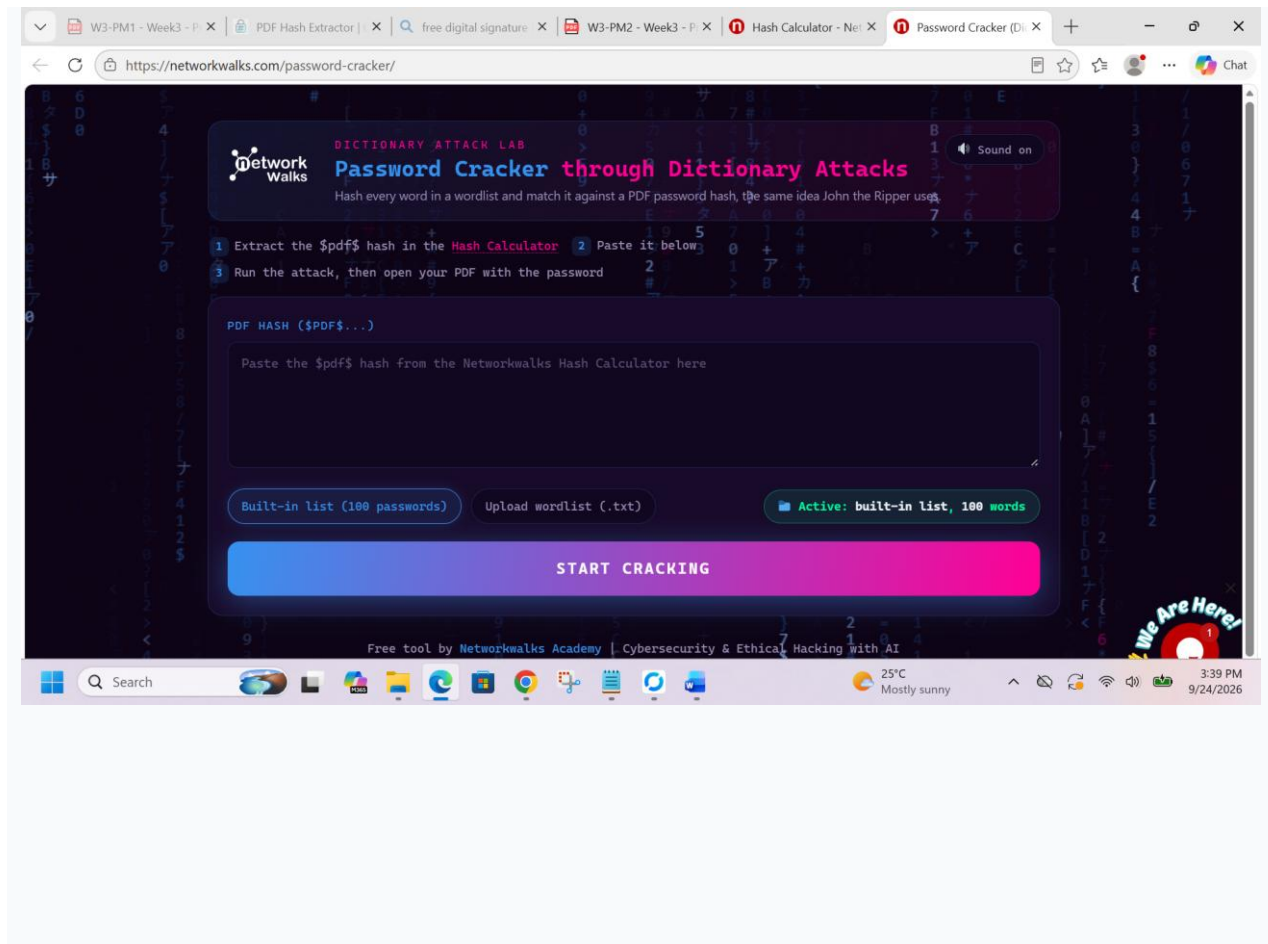


Figure 5. Networkwalks Password Cracker.

STEP 6 | Submit the hash and start the attack

Paste the extracted PDF hash into the Password Cracker and start the attack. The lab describes the process as trying different passwords until a matching value is found.

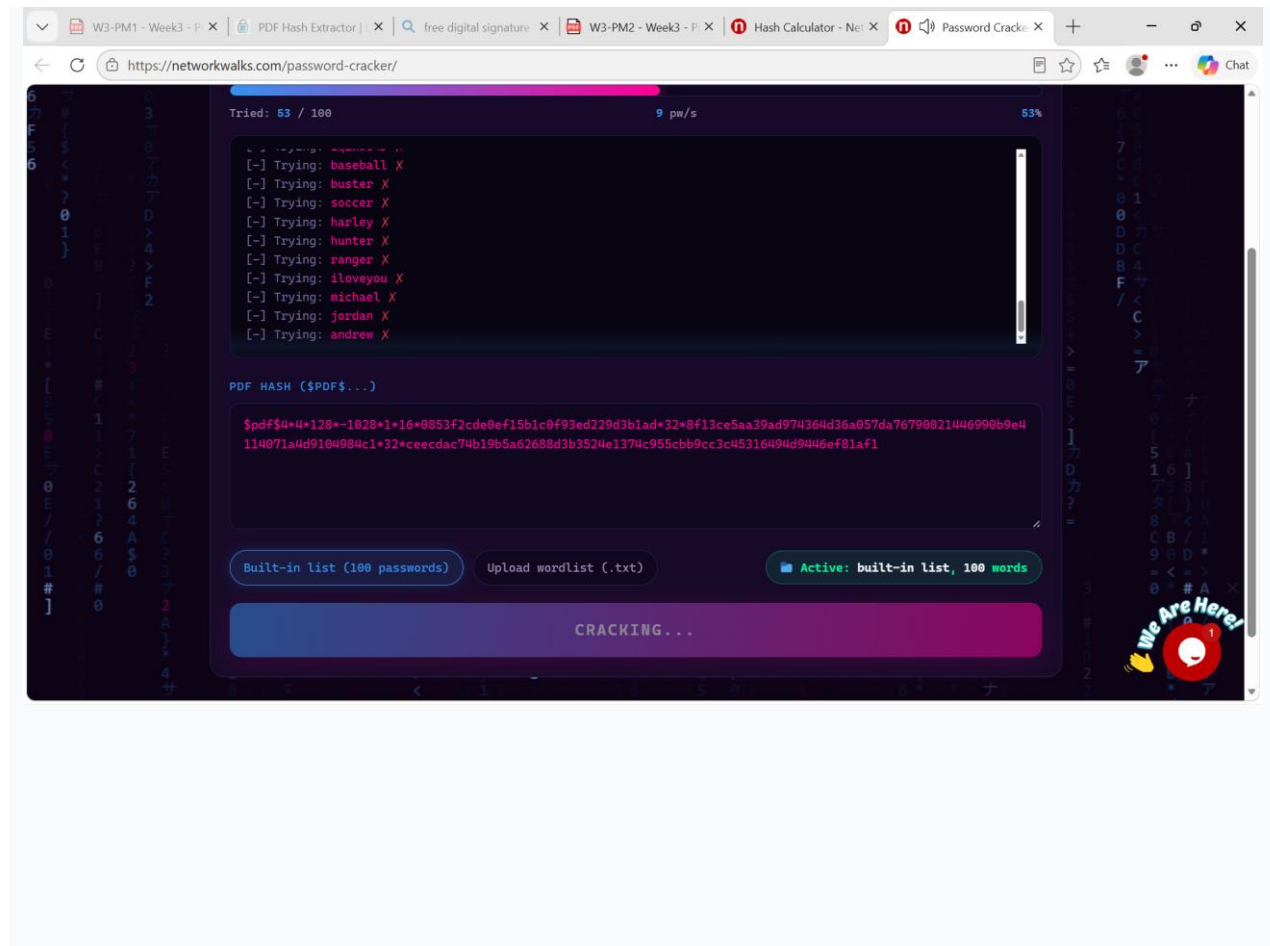


Figure 6. Cracking process started.

STEP 7 | Record the recovered password

Wait for the Password Cracker to complete. The supplied practical states that the recovered password is displayed on the screen and that the time required depends on the simplicity or complexity of the password.

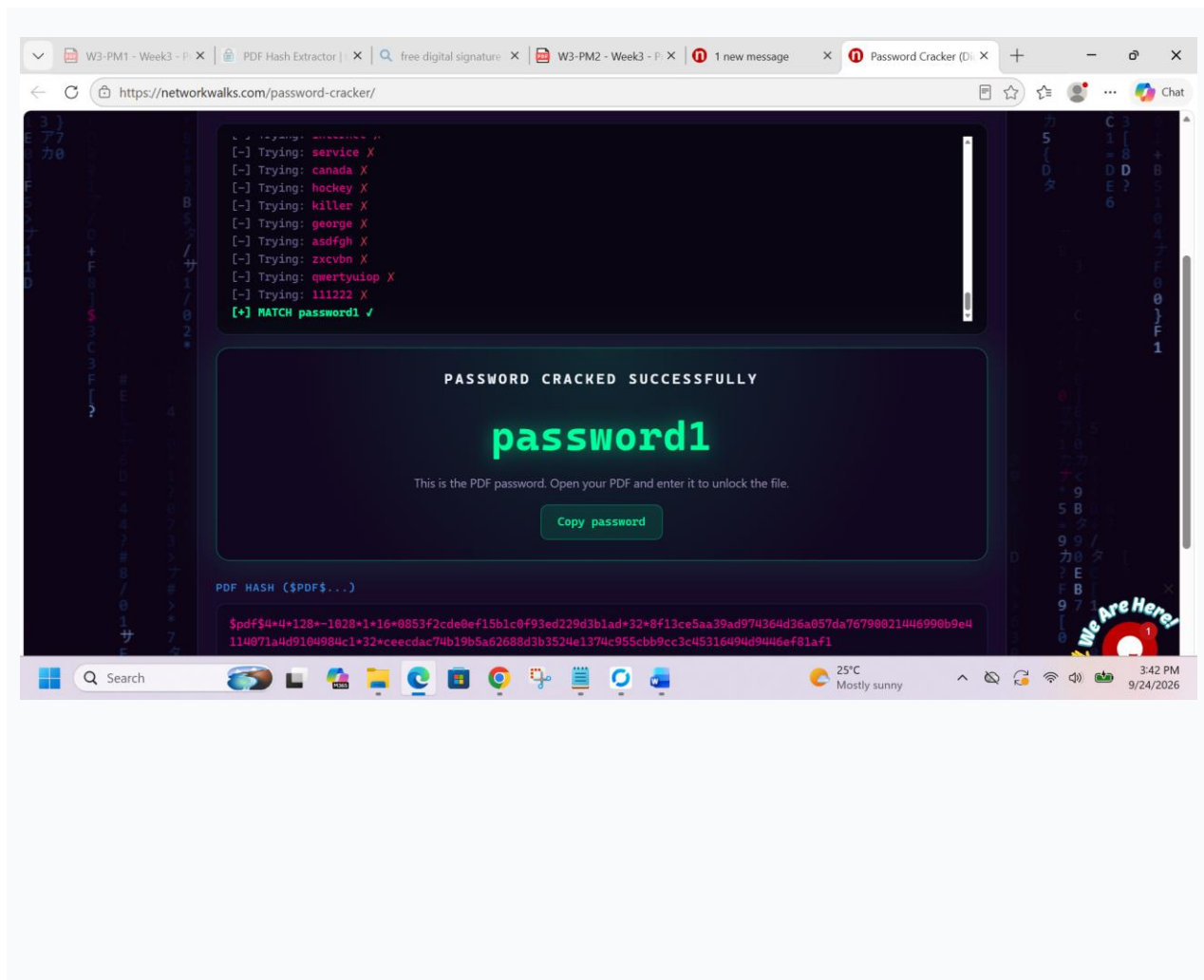


Figure 7. Password recovered.

STEP 8 | Enter the recovered password into the PDF

Open the locked PDF in a PDF reader and enter the recovered password. The example in the supplied task document shows the password “password1”. For your own evidence, record the password recovered during your practical below.

Recovered password: password1

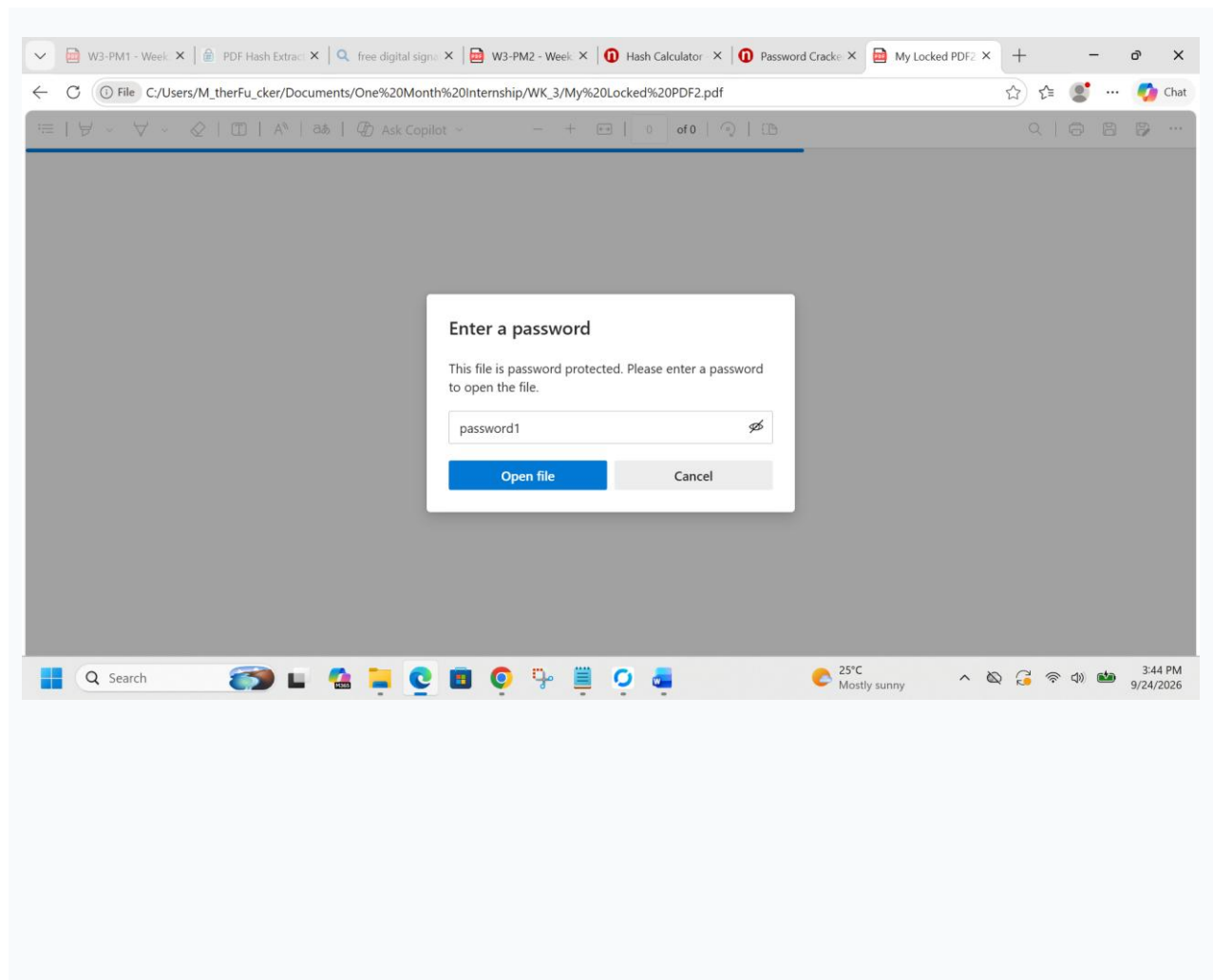


Figure 8. Password entered into the PDF.

STEP 9 | Confirm successful access

After the password is accepted, confirm that the PDF opens successfully. The task document identifies successful opening of the protected PDF as the completion point of the lab.

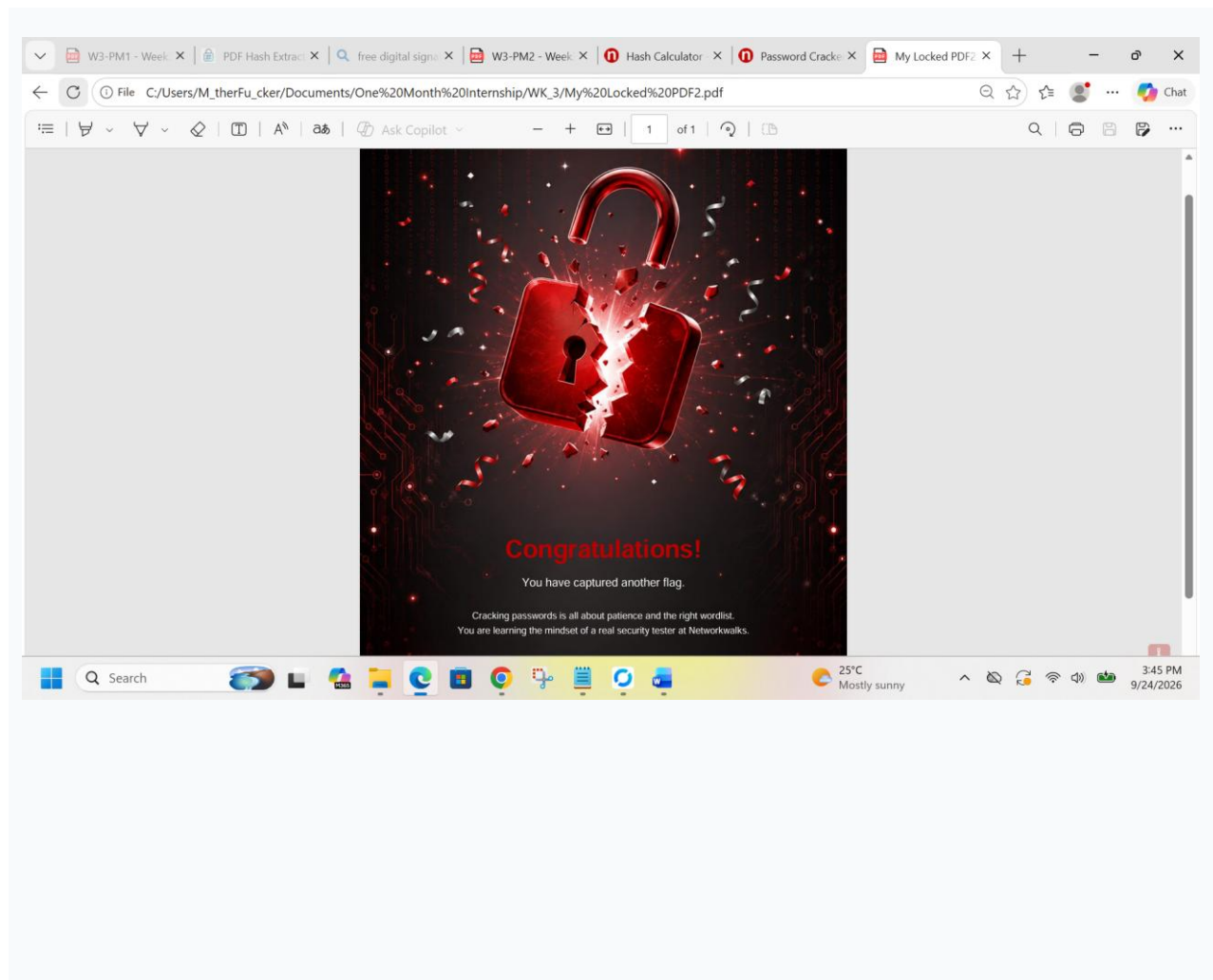


Figure 9. Locked PDF opened successfully.

7. Results

The practical should be considered successfully completed when the recovered password is accepted by the protected PDF and the document opens. Complete the table below after carrying out the task.

PDF file	My Locked PDF2
Hash extracted	\$pdf\$4*4*128*- 1028*1*16*0853f2cde0ef15b1c0f93
Cracking status	Successful
Recovered password	password1
PDF access confirmed	Yes
Completion date	24/09/2026

8. Evidence Checklist

- ✓ Locked PDF downloaded
- ✓ Hash Calculator opened
- ✓ PDF uploaded to Hash Calculator
- ✓ \$pdf\$ hash extracted
- ✓ Complete hash copied
- ✓ Password Cracker opened
- ✓ Hash submitted and attack started
- ✓ Password recovered
- ✓ Password entered into locked PDF
- ✓ PDF opened successfully

9. Discussion

This exercise demonstrates the relationship between a protected file, a password-derived hash, and password recovery attempts. The practical illustrates that a password that appears sufficient to protect a file can still be vulnerable when it is simple or predictable enough to be identified through systematic password testing.

The supplied lab material also distinguishes encryption from hashing. It describes encryption as a two-way process in which protected information can be recovered using the appropriate key, whereas hashing is described as a one-way process that transforms plaintext into a message digest. For security practice, the exercise reinforces the importance of using strong, unique passwords rather than short or commonly used passwords.

10. Ethical and Security Considerations

Password-cracking techniques should only be used on systems, files, accounts, or datasets for which the tester has explicit authorization. In this practical, the activity is framed as a controlled cybersecurity training exercise using the supplied laboratory file. The purpose is to understand password security and demonstrate the risks associated with weak credentials, rather than to gain unauthorized access to third-party information.

11. Conclusion

The Week 3 Project Module 2 task provides a practical demonstration of password recovery using the Networkwalks Hash Calculator and Password Cracker. The process consists of extracting the PDF hash, submitting the hash to the cracking tool, recovering the password, and using that password to open the protected PDF. The completed screenshots and results table in this report provide evidence of the practical work performed.

12. References

- Networkwalks. Week 3 | Project Module 2: Password Cracking with Networkwalks Tools. Supplied practical PDF.
- Networkwalks Hash Calculator: <https://networkwalks.com/hash-calculator/>
- Networkwalks Password Cracker: <https://networkwalks.com/password-cracker/>
- Networkwalks Project Task Page: <https://networkwalks.com/project-task-lab-password-cracking-with-networkwalks-toolsa/>

Learner Declaration

I confirm that the practical work documented in this report was performed in an authorized training/lab environment and that the screenshots included in this report represent my own practical evidence.

Name: ASANDA LLOYD NYUSWA Signature:

Date: 24/09/2026